In [1]:

```
from collections import Counter
import nltk
nltk.download('punkt')
nltk.download('punkt_tab')

# Sample text corpus
text = "this is a sample text this is simple"

# Tokenization
tokens = nltk.word_tokenize(text)

# ---------------- UNIGRAM ----------------
unigram_counts = Counter(tokens)
total_words = len(tokens)
vocab_size = len(unigram_counts)

print("Unigram Probabilities (Add-One Smoothing):")
for word in unigram_counts:
    prob = (unigram_counts[word] + 1) / (total_words + vocab_size)
    print(word, ":", prob)

# ---------------- BIGRAM ----------------
bigrams = list(nltk.bigrams(tokens))
bigram_counts = Counter(bigrams)

print("\nBigram Probabilities (Add-One Smoothing):")
for bg in bigram_counts:
    prob = (bigram_counts[bg] + 1) / (unigram_counts[bg[0]] + vocab_size)
    print(bg, ":", prob)

# ---------------- Sentence Probability ----------------
sentence = "this is simple"
sentence_tokens = nltk.word_tokenize(sentence)

probability = 1
for i in range(len(sentence_tokens)-1):
    bg = (sentence_tokens[i], sentence_tokens[i+1])
    prob = (bigram_counts[bg] + 1) / (unigram_counts[sentence_tokens[i]] + vocab
    probability *= prob

print("\nSentence Probability:", probability)
```

```
[nltk_data] Downloading package punkt to
[nltk_data]     C:\Users\Priyansh\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to
[nltk_data]     C:\Users\Priyansh\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
```

```
Unigram Probabilities (Add-One Smoothing):
this : 0.21428571428571427
is : 0.21428571428571427
a : 0.14285714285714285
sample : 0.14285714285714285
text : 0.14285714285714285
simple : 0.14285714285714285

Bigram Probabilities (Add-One Smoothing):
('this', 'is') : 0.375
('is', 'a') : 0.25
('a', 'sample') : 0.2857142857142857
('sample', 'text') : 0.2857142857142857
('text', 'this') : 0.2857142857142857
('is', 'simple') : 0.25

Sentence Probability: 0.09375
```

In [ ]: